

WEEK-10--Elasticbeanstalk

Prerequisites

- A web application ready (e.g., .war, .zip, or code)
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Steps to Create Elastic Beanstalk Application

Step 1: Open Elastic Beanstalk

1. Login to AWS Console
 2. Under Compute section ->Search for **Elastic Beanstalk**
 3. Click **Create Application**
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Step 2: Create Application

1. Enter **Application Name** (e.g., MyApp)
 2. Add description (optional)
 3. Click **Create**
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Step 3: Create Environment

1. Click **Create Environment**
 2. Choose:
 - **Web Server Environment** (for web apps)
 3. Click **Select**
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Step 4: Configure Environment

1. Enter **Environment Name**
2. Choose **Platform**:
 - tomcat (for .war)
 - tomcat /Node.js / Python / PHP / etc.
3. Under **Application Code**:

- Select **Upload your code**
 - Upload your .war or .zip file
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Step 5: Configure Service Access

1. Service Role → Select existing -> **LabRole**
 2. EC2 Key Pair → Optional (for SSH)
 3. EC2 Instance Profile → select (e.g., LabInstanceProfile)
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Step 6: Configure Network

1. Select **VPC** (default is fine)
 2. Select **Subnets**
 3. Enable **Public IP**
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Step 7: Instance & Scaling Setup

1. Choose **Instance type** (e.g., t2.micro – free tier)
 2. Set:
 - Min instances: 1
 - Max instances: 2 (or more)
 3. Load Balancer → Enabled (optional but recommended)
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Step 8: Review & Create

1. Click **Review**
 2. Click **Create**
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Deployment Process

- AWS automatically creates:
 - EC2 instances
 - Load Balancer

- Auto Scaling Group
 - Security Groups
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Access Your Application

- After deployment:
 - You will get a **URL (domain link)**
 - Open it in browser → Your app will run
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Updating Application

1. Go to your environment
 2. Click **Upload and Deploy**
 3. Upload new version
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Scenario-Based Questions

Scenario 1: Application Not Loading

You deployed an application using Elastic Beanstalk, but the application does not load and shows an error page.

- What could be the possible reasons for this issue?
 - Could deployment or configuration be incorrect?
 - How would you troubleshoot this problem?
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Scenario 2: Incorrect Application Structure

Your application is deployed, but the expected output is not displayed.

- How does application structure affect deployment in Elastic Beanstalk?
- What could happen if files are placed inside an extra folder?
- How would you fix the structure issue?

Scenario 3: Wrong Platform Selection

While creating the environment, an incorrect platform was selected.

- Why is platform selection important in Elastic Beanstalk?
- What kind of errors or behavior might occur?
- How can you correct the platform without affecting the application?

Scenario 4: Environment Health Warning

The environment status shows Warning (Yellow) instead of healthy.

- What are possible reasons for this warning?
- Which sections (Events, Logs, Metrics) would you check first?
- How can you resolve health issues?

Scenario 5: Application Errors in Logs

Logs show a high number of application errors.

- What do such errors indicate?
- Could this be due to misconfiguration or missing resources?
- How would you analyze logs to identify the root cause?

Scenario 6: Deployment Failure

The application deployment fails during upload.

- What could be wrong with the deployment package?
 - Which logs would you check to diagnose the issue?
 - How can packaging errors affect deployment?
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Scenario 7: Works Locally but Not on Cloud

The application works correctly on a local machine but fails after deployment.

- What environment differences could cause this issue?
 - Could configuration or dependency issues be responsible?
 - How would you debug using cloud logs?
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Scenario 8: Blank Output

The application loads but shows no output.

- What could cause an application to display no output?
 - Could runtime errors or misconfigurations be responsible?
 - How would logs help in identifying the issue?
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Scenario 9: Application URL Not Accessible

The application URL is not accessible after deployment.

- Could the environment still be launching?
 - What status indicators should be checked?
 - How can Events help identify the issue?
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Scenario 10: Resource Management Issue

After completing the lab, resources were not terminated.

- What are the cost implications of leaving resources running?
 - Which AWS services continue running in the background?
 - How can all resources be properly terminated?
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Scenario 11: High Traffic Handling

The application receives sudden high traffic.

- How does Elastic Beanstalk handle scaling?
 - What configurations should be enabled?
 - What happens if scaling is not configured?
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Scenario 12: Application Update Not Reflected

Changes are made to the application, but users still see the old version.

- Could deployment issues be the reason?
- How can caching affect the output?
- How would you ensure the latest version is visible?